

DEBUGGING A NOTE TAKING APPLICATION

> PROBLEM STATEMENT

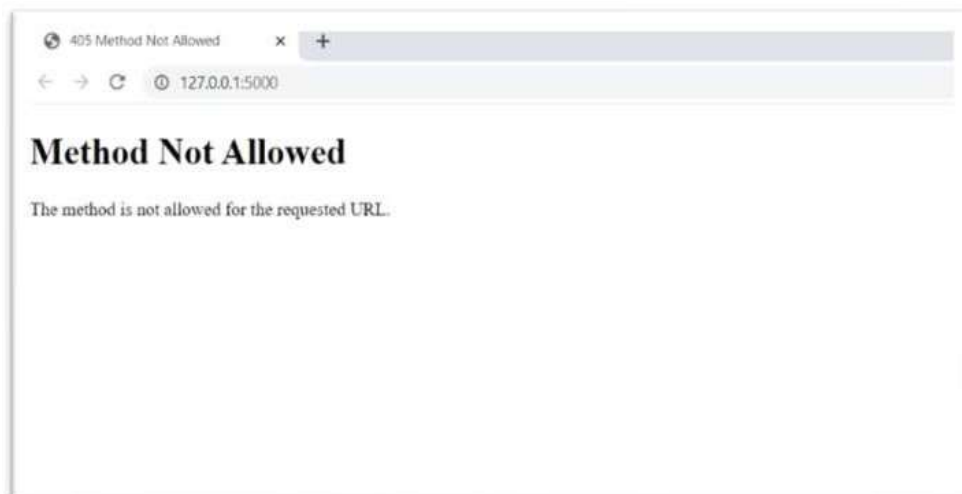
Given a note taking application having one route (i.e., Home Route). This application is broken.

Home Route contains a 'text field' and a 'button'. Users can add a note and all the notes will be displayed as an unordered list below the text field on the same page.

- We have to refactor the code in order to work the application properly by debugging the bugs and fixing them.
- This report contains the details of all the bugs in the application and the approach adopted to resolve all the bugs.

BUG 1: Mistake in Request Method used

The first bug was that the method for default route "/" was receiving POST request when the actual requests sent by the form was GET request.



Solution:

1. Changed the arguments receiving method in index () from **args.get ()** to **form []**.

BEFORE

AFTER

```

home copy1.html  app copy1.py
Innomatics > m_asgmt_debug > app copy1.py > ...
1  from flask import Flask, render_template, request
2  app = Flask(__name__)
3
4
5  notes = []
6  @app.route('/', methods=["POST"])
7  def note_func():
8      note = request.args.get('note')
9      notes.append(note)
10     return render_template('home.html', notes=notes)
11
12
13 if __name__ == '__main__':
14     app.run(debug=True)
15

```

```

home copy1.html  app copy1.py
Innomatics > m_asgmt_debug > app copy1.py > ...
1  from flask import Flask, render_template, request
2  app = Flask(__name__)
3
4
5  notes = []
6  @app.route('/', methods=["POST", "GET"])
7  def note_func():
8      if request.method == "POST":
9          note = request.form['note']
10         notes.append(note)
11         return render_template('home.html', notes=notes)
12     else:
13         return render_template('home.html')
14
15
16 if __name__ == '__main__':
17     app.run(debug=True)
18

```

2. Changed the method by which data was sent from html file from **GET** to **POST**.

BEFORE

AFTER

```

home copy1.html X  app copy1.py
Innomatics > m_asgmt_debug > templates > home copy1.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta http-equiv="X-UA-Compatible" content="IE=edge">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Add Note</title>
8      <style>
9          h1 {text-align: center;}
10         h2 {text-align: center;}
11         input[type="text"]{display: block; margin: 0 auto; padding: 10px}
12         button{display: block; margin: 0 auto}
13     </style>
14     <h1>BACKEND PROJECT 2</h1>
15 </head>
16 <body>
17     <h2><b>Debugging a Note Taking Application</b></h2><hr>
18     <form action="">
19         <input type="text" name="note" placeholder="Enter a note"><br>
20         <button>Add Note</button><hr>
21     </form>
22     <h3>Added Notes :</h3>
23     <ul>
24         {% for note in notes %}
25         <li>{{ note }}</li>
26         {% endfor %}
27     </ul>
28 </body>
29 </html>

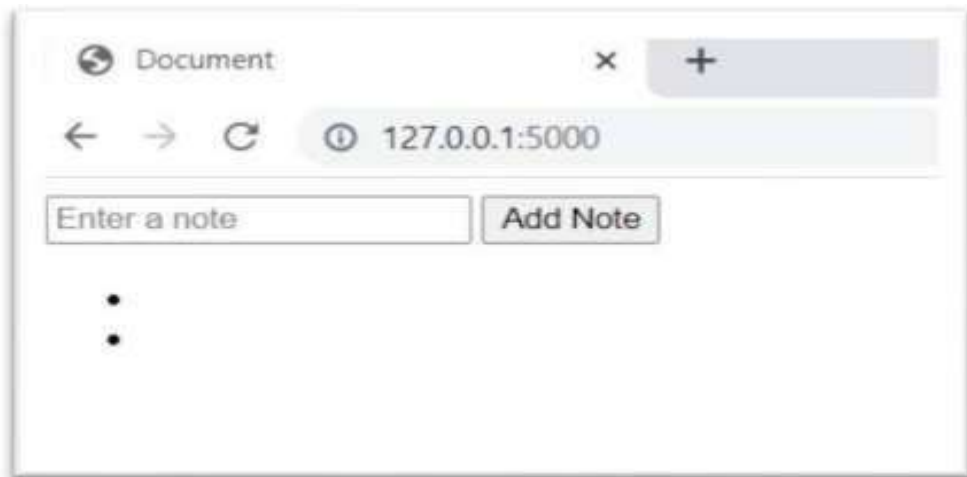
```

```

home copy1.html  app copy1.py
Innomatics > m_asgmt_debug > templates > home copy1.html > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta http-equiv="X-UA-Compatible" content="IE=edge">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Add Note</title>
8      <style>
9          h1 {text-align: center;}
10         h2 {text-align: center;}
11         input[type="text"]{display: block; margin: 0 auto; padding: 10px}
12         button{display: block; margin: 0 auto}
13     </style>
14     <h1>BACKEND PROJECT 2</h1>
15 </head>
16 <body>
17     <h2><b>Debugging a Note Taking Application</b></h2><hr>
18     <form action="" method = 'POST'>
19         <input type="text" name="note" placeholder="Enter a note"><br>
20         <button>Add Note</button><hr>
21     </form>
22     <h3>Added Notes :</h3>
23     <ul>
24         {% for note in notes %}
25         <li>{{ note }}</li>
26         {% endfor %}
27     </ul>
28 </body>
29 </html>

```

BUG 2: Empty String / None is added as Note



When add note is pressed after typing some content into the text box, the app added None/Empty String as a note.

Solution:

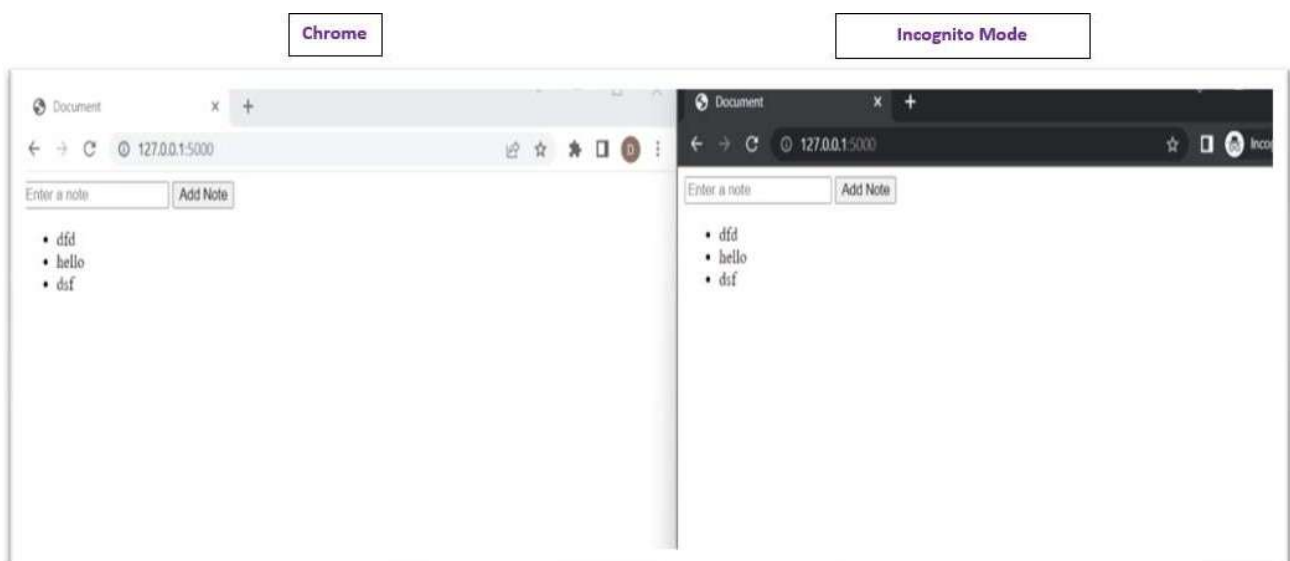
- 1) When we change **GET method to POST method** the problem of adding None as a note is resolved.
 - When Add Note button is pressed without passing any argument into the text box, `args.get ()` method takes None as its type by default if GET method is used.
 - When we use POST method instead of GET method, `args.get ()` method does not take default arguments but shows empty strings.
- 2) To resolve the problem of adding Empty String as a note, we use **If condition** for checking the string entered in text box is empty or not.
 - If an empty string is entered in text box, then it should not be added to notes.

```
app copy2.py x
Innomatics > m_asgmt_debug > app copy2.py > ...
1  from flask import Flask, render_template, request
2  app = Flask(__name__)
3
4
5  notes = []
6  @app.route('/', methods=["POST", "GET"])
7  def note_func():
8      if request.method == "POST":
9          note = request.form['note']
10         if note!="":
11             notes.append(note)
12             return render_template("home.html", notes=notes)
13         return render_template('home.html')
14
15
16 if __name__ == '__main__':
17     app.run(debug=True)
18
```

BUG 3: Sessions not added

When two browsing windows (One window as **Chrome** and other window as **Incognito mode**) are running parallel on the same app, the content of notes added in one window session is displayed in another window also.

- This does not make the application platform-independent which cause security risks.



Solution:

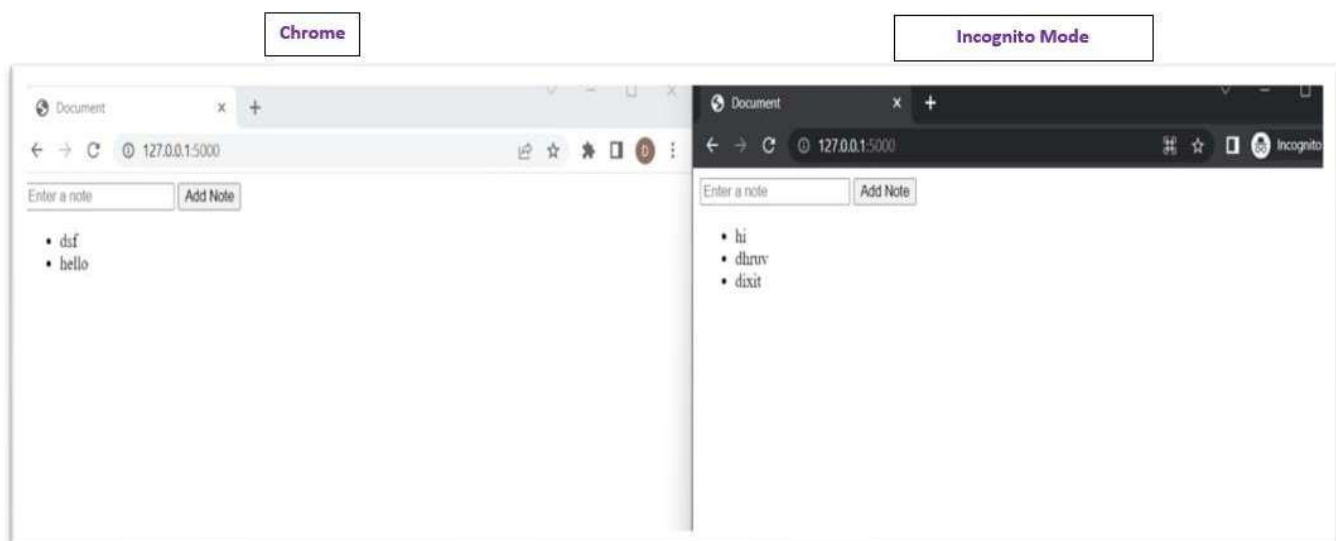
To resolve this problem, we can use **Flask-Session**.

- Flask-Session is an extension for Flask that supports **Server-side Session** to your application.
- The Session is the **time between** the **client logs in** to the server and **logs out** of the server.
- **Each client** will have their **own session** where their own data will be stored in their session.

Code implemented with Flask-Session:

```
app.py x
Innomatics > m_asgmt_debug > app.py > ...
1 from flask import Flask, render_template, request, session
2 from flask_session import Session
3
4 app = Flask(__name__)
5 app.config["SESSION_PERMANENT"] = False
6 app.config["SESSION_TYPE"] = "filesystem"
7 Session(app)
8
9 notes = []
10 @app.route('/', methods=["POST", "GET"])
11 def note_func():
12     if session.get("notes") is None:
13         session["notes"] = []
14     if request.method == "POST":
15         note = request.form["note"]
16         if note != "":
17             session["notes"].append(note)
18     return render_template("home.html", notes=session["notes"])
19
20 if __name__ == '__main__':
21     app.run(debug=True)
22
```

Now at-least 2-3 users can use the same app at same time on the same machine as follows:



NOTE TAKING APPLICATION

← → ↺ ☆ | http://127.0.0.1:5000 / Notes App

🔍 📄 📁 ⋮

 **NoteVault**
Lock in your notes !!

Enter title

Enter your note here...

Save Note

Sort: Latest First ▾

📌 S

Latest First

Oldest First

Title (A-Z)

es

Trip Necessities Checklist

EditDelete

Product Sync – Jan 16

EditDelete

Photosynthesis

EditDelete

Clear All Notes

Trip Necessities Checklist

Documents Required

- * ID / Passport
- * Visa (if required)
- * Tickets (flight/train/bus)
- * Hotel booking confirmation
- * Travel insurance
- * Driver's license

Edit

Delete

Back